

## Scalability & Future Plan

|               |                                                                      |
|---------------|----------------------------------------------------------------------|
| Date          | 03 July 2026                                                         |
| Team ID       |                                                                      |
| Team Size     | 3                                                                    |
| Team Leader   | Chakali Basavaraju                                                   |
| Team Member 1 | Girish C                                                             |
| Team Member 2 | Shaik Sameer                                                         |
| Team Member 3 |                                                                      |
| Team Member 4 |                                                                      |
| Project Name  | Smart Lender: Machine Learning-Based Loan Approval Prediction System |
| Maximum Marks |                                                                      |

### Scalability & Future Plan;

This document outlines how the current **Smart Lender Loan Approval Prediction System** can be scaled to support larger user bases, increased data volumes, and advanced feature enhancements. It also defines a roadmap for evolving the system into a more robust, scalable, and intelligent financial solution.

|                                  | Planned Feature                                                 | Target Timeline | Expected Impact                                                                |
|----------------------------------|-----------------------------------------------------------------|-----------------|--------------------------------------------------------------------------------|
| phase t<br>(Short-tenn: "-6 mos) | Automated Model Retraining<br>Hyperparameter Tunang<br>Pipeline | 03 Jan 2027     | Maintained model performance over time<br>and automated Contious Optimization. |
| Phase 2<br>(Mid-term •12mos)     | Feature Enrichment (Interactipon<br>and demographic features)   | 03 July 2027    | Improved model accuracy by 5-8% on<br>edge case predictions.                   |
| phase 3 term:<br>mos)            | Explainable AI (XAI)<br>Dashboard foradministrators             | 03 Jan 2028     | Increased transparency in prediction<br>decisions and enabling faimess         |

### Current System Limitations

| S.No | Limitation                                                                                  | Impact                                                                    | priority to Address<br>(High Medium / Low) |
|------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|--------------------------------------------|
| 1    | Limited historical data range (e.g., only 2 years).                                         | Reduced model training dataset diversity and potential for temporal bias. | Medium                                     |
| 2    | High prediction latency during peak usage hours (single worker).                            | Slower response time and possible request timeout                         | High                                       |
| 3    | Basic feature engineering (missing advanced financial indicators like debt-to-income ratio) | Reduced model accuracy in edge cases                                      | Medium                                     |

Scalability Plan:

| S.No | Scalability Aspect | Current State                                           | Proposed Upgrade / Solution                                                            |
|------|--------------------|---------------------------------------------------------|----------------------------------------------------------------------------------------|
| 1    | User Load          | Single-instance, synchronous Flask application.         | Use Gunicorn with multiple workers and enable auto-scaling via Docker-based deployment |
| 2    | Data Storage       | Single, non-replicated local MySQL instance.            | Migrate to managed MySQL cluster with read replicas for high availability              |
|      | Performance        | Model inference executes on the main application thread | Use asynchronous task queues (e.g., Celery) and optimize model using ONNX              |
| 4    | Security           | Basic API key authentication with unencrypted storage.  | Implement OAuth 2.0, JWT authentication, RBAC, and database encryption                 |

Future Roadmap:

Scalability & Future Plan

| Scalability Area / Phase | Proposed Upgrade / Enhancement                                                                                                                                                                                     | Expected Performance Impact                                                                                                                                                                | Target Timeline |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Phase 2                  | Deploy system on autoscaling cloud infrastructure (e.g., AWS ECS with Fargate) and database replication.                                                                                                           | Improved performance, higher concurrent user support, reduced latency                                                                                                                      | Q1 2027         |
| Phase 3                  | Integrate continuous model monitoring and drift detection APIs. Explore integration with advanced anomaly detection models or third-party datasets for enhanced prediction features.                               | Proactive identification of model performance issues.<br><br>Potentially higher prediction accuracy.<br>Expanded system capabilities.                                                      | Q3 2027         |
| Phase 4                  | Hardening the system security layer.<br><br>Developing a secured mobile application interface.<br><br>Expanding model applicability to new related product domains.<br><br>Implement CI/CD for automated pipeline. | Reduced vulnerability to cyber-attacks.<br><br>Increased market reach and accessibility.<br><br>Higher overall system reliability.<br><br>Enhanced business value and strategic alignment. | Q1 2028         |